William Liu

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Experience

Meta Jun 2026 – Aug 2026

Software Engineer Intern *Menlo Park, CA*

- Deployed an automated service running **3,300+** weekly failover readiness checks across Meta's edge network, protecting user traffic by proactively preventing reduced video and media quality.
- Cut network incident assessment time to <**10 minutes** by building automated user impact analyzers for egress and goodput.
- Surfaced **130+** previously unexamined network failover incidents by designing and deploying a real-time Python service that continuously detects organically rerouted traffic events.
- Built a network degradation report dashboard to help drive capacity planning decisions with ISP partners and minimize outages.

Major League Hacking Jun 2025 – Sep 2025

Site Reliability Engineer (SRE) Fellow *Remote*

- Reduced the deployment time of a Flask web app by **41%** by optimizing Docker containers through layer caching and automating releases with GitHub Actions and Bash.
- Proactively monitored Docker and Linux resources with Prometheus and Grafana, alerting on resource spikes within **2 minutes**.

Trojan Technologies May 2025 – Aug 2025

Software Engineer Intern *London, ON*

- Enabled data teams to build analytics and AI models on UV system telemetry by migrating **27%** more legacy tables from Azure to Snowflake through automated Python and Airflow pipelines.
- Raised pipeline backfilling coverage by **53%** to eliminate gaps in historical device data caused by pipeline ingestion failures for an IoT monitoring platform, increasing the accuracy of customer-facing data.
- Increased ETL pipeline release frequency from **weekly to daily** by building a Jenkins CI/CD pipeline with Bash automation.

Education

Western University **Expected Graduation:** June 2027

Honors in Computer Science and Honors in Business Administration (HBSc. + HBA) *GPA: 3.8/4.0*

- Awards: Dean's Honor List, Western Scholarship of Excellence ($8,000), Andrew & Sarah Hamilton Scholarship ($6,600)
- Relevant Coursework: Data Structures & Algorithms, Software Tools & Systems Programming, Computer Organization & Architecture, Operating Systems, Object-Oriented Design & Analysis

Leadership

Hack Western May 2023 – Jan 2026

Lead Web Developer *London, ON*

- Architected the full-stack platform (frontend, backend, database, deployment) for one of Canada's largest collegiate hackathons, powering promotion, applications, and event-day operations for **1,400+** applicants and **300+** accepted participants.
- Led 6 developers through feature planning, code reviews, and task delegation to ship the hackathon's web platform in 8 months.
- Engineered a React-based draggable canvas and interactive elements animated using CSS for hackathon promotional website.

Projects

Anishelf | *TypeScript, Next.js, Flask, PostgreSQL, GraphQL, Docker, nginx*

- Developed a full-stack app using Next.js & Flask that parses TV subtitles to auto-generate frequency-ranked vocabulary flashcards for foreign language learners in under 10 seconds.
- Deployed app using Docker containers on a DigitalOcean VPS, utilizing nginx as a reverse proxy to manage traffic.

YACHIP-8 Emulator | *Rust, SDL2*

- Built a CHIP-8 emulator in Rust, implementing the CPU instruction set, registers, memory, and stack to run unmodified ROMs.
- Developed an SDL2 front end rendering the 64x32 display with keyboard input mapping and timer-accurate execution at 60Hz.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, Rust, Go, C++, Bash, HTML, CSS
Frameworks/Libraries: React, Node.js, Express.js, Next.js, Flask, SQLAlchemy, Selenium, Spring Boot, JUnit
Tools: Git, Linux, Docker, Kubernetes, MongoDB, PostgreSQL, MySQL, GraphQL, AWS, DigitalOcean, nginx, Jenkins